

2018-Ph-H1-Q8

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

A car accelerates from rest at 3.0 m/s^2 for 8.0 s .

What distance does it travel?

Solution:

Use:

$$s = ut + \frac{1}{2}at^2$$

$$s = 0 + \frac{1}{2} \times 3.0 \times 8.0^2 = \frac{3.0}{2} \times 64 = 96 \text{ m}$$

Answer: 96 m

Revision Tip:

This is a classic SUVAT situation: starts from rest (so $u = 0$), and you're given a and t . Pick the formula that matches.